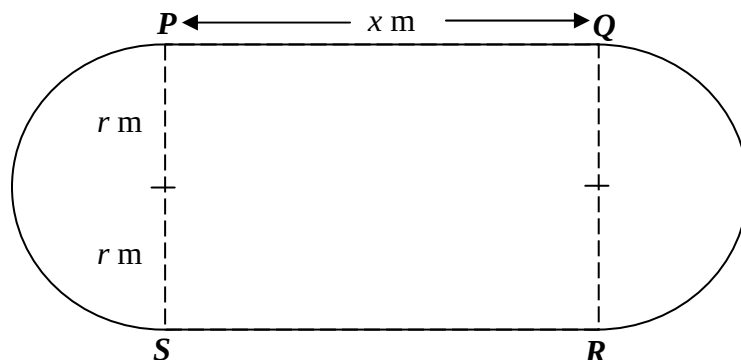


Section A: Pure Mathematics [40 marks]

- 1** Given that $\log_3 2 = p$ and $\log_3 5 = q$, express in terms of p and q ,
- (i) $\log_3 20$,
 - (ii) $\log_{10} 5$. **[4]**
- 2** The curves $y = 3x^2 - 2x + 1$ and $y = x^3 - ax^2 + x + \frac{2}{3}b$ have a common turning point. Find
- (i) the coordinates of the turning point, **[3]**
 - (ii) the exact values of a and b . **[4]**
- 3** In the same diagram, sketch the graphs of $y = 3e^{\frac{x}{3}}$ and $y = \frac{3}{(x+1)^2}$. Show on your diagram, the equations of the asymptotes and the coordinates of the points of intersection of the two graphs. Find the area of the region bounded by the 2 graphs and the line $x = 2$. **[5]**
- 4** (a) Find $\frac{d}{dx}(e^{x^2-1})$. Hence find $\int xe^{x^2-1} dx$. **[3]**
- (b) Evaluate $\int_1^4 \frac{xe^x - 3 + 2\sqrt{x}}{x} dx$, giving your answer in the exact form. **[4]**
- 5.** The functions f and g are defined, for all real values of x by
- $$f : x \rightarrow 1 - 2x^2,$$
- $$g : x \rightarrow \ln(2x - 1).$$
- (i) On separate diagrams, sketch the graphs of f and g . **[4]**
 - (ii) State the ranges of f and g . **[2]**
 - (iii) Explain clearly why f^{-1} does not exist. Write down the minimum value of α such that f^{-1} exists when the domain of f is restricted to (α, ∞) . **[2]**
 - (iv) Find $g^{-1}(x)$ and state its domain. **[2]**

- 6 The figure shows the outline of a proposed 400m running track for a school, which enclose a rectangular region $PQRS$ and a semicircular region at each end, with PS and QR as diameters where $PS = QR = 2r$ m.



- (a) Given that $PQ = x$ m, show that the area, A , enclosed by the running track is given by $A = 400r - \pi r^2$. [3]
- (b) Suppose the management wants to maximise the area enclosed by the 400m track by adjusting the radius for the semicircular regions. Find the maximum area enclosed as r varies. [4]

Section B: Statistics [60 marks]

- 7 A college has 2400 students. A student wishes to take a sample of size 120 to survey the mode of transport his fellow college mates take to college.
- (i) Describe how the student could select a systematic sample of the students. [2]
- (ii) State one advantage and one disadvantage of this method of sampling as opposed to conducting a survey of the entire population of students in the college. [2]

The results of the survey were as follows:

Mode of transport	Bus	Car	Bicycle	Walk
Number of students	45	60	5	10

- (iii) Use the sample information to estimate the number of students in the college who walk to college. [2]

- 8 A new index on overall intelligence of 10-year old children is being developed. The index, x , is measured for a random sample of 150 children of age 10 years. The results are summarized by

$$\sum x = 7170, \quad \sum x^2 = 364182.$$

Find unbiased estimates of the population mean and variance. [3]

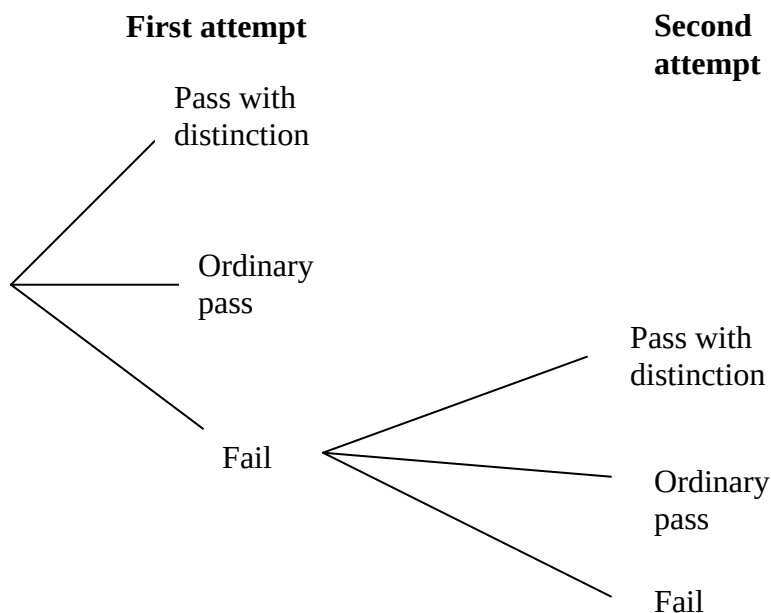
Test, at the 5% significance level, whether the population mean index for a 10-year child is less than 50. [4]

State, with a reason, whether, in using the above test, it is necessary to assume that the index on overall intelligence of 10-year old children have a normal distribution. [1]

- 9 A music school requires candidates to take the practical examination a second time if they fail it on the first attempt.

For those taking the examination on the first attempt, 25% pass with distinction and 50% gain an ordinary pass. For those attempting the examination a second time, 5% pass with distinction and 60% gain an ordinary pass. The tree diagram below illustrates the situation.

/



- (a) Find the probability that a randomly chosen candidate
- (i) fails the examination, [2]
- (ii) passes the examination (with or without distinction). [1]
- (b) Given that a randomly chosen candidate passes the examination, find the probability that he or she passes with distinction. [3]
- (c) Kelly and Jane are two randomly chosen candidates for the examination. Find the probability that Kelly gets a better result than Jane. [4]

- 10** In a long term study, it has been found that the proportion of a certain type of rice seed that will germinate is 35%.
- (a)** Ahmad plants 30 such seeds at the beginning of the growing season in two batches, each of 15 seeds. Find the probability that
- (i)** exactly 10 of the 30 seeds germinate, [2]
- (ii)** each batch produces at most 7 germinating seeds. [4]
- (b)** Ali plants 300 of such seeds at the beginning of the growing season. Using a suitable approximation, find the probability that at least 120 seeds germinate. [4]
- 11** A supermarket takes delivery of bags of potatoes which have a normal distribution with mean 5.5 kg and standard deviation σ kg. If 10% of the bags of potatoes weigh less than 5 kg, show that the standard deviation is 0.4 kg correct to 1 s.f. [4]
- (i)** A customer randomly chooses 2 bags of potatoes. Find the probability that the total mass of the 2 bags of potatoes is at least 10 kg. [3]
- (ii)** A bag of potatoes is sold at \$2 per kg. Find the probability that a randomly chosen bag of potatoes has a selling price exceeding \$12. [3]
- (iii)** A random sample of n bags of potatoes is taken. Given that the probability that the sample mean exceeds 5.6 kg is at least 0.2, find the set of possible values of n . [4]
- 12** The *data* in the table below shows the ages in years (x) and the trunk circumferences in cm (y) of a random sample of 7 trees of a particular species.

Age (x)	11	13	21	28	34	45	51
Circumference (y)	24.4	32.1	52.8	78.3	79.2	102.7	121.2

- (i)** Give a sketch of the scatter diagram for the data, as shown on your calculator. [2]
- (ii)** Calculate the product moment correlation coefficient. [1]
- You are given that the regression line of y on x has equation $y = 3.21 + 2.31x$, where the coefficients are given correct to 3 significant figures.
- (iii)** Calculate the equation of the regression line of x on y , giving your answer in the form $x = a + by$. [1]
- (iv)** Use the appropriate regression line to estimate,
- (a)** the circumference of the trunk of a tree which is 25 years old, [2]
- (b)** the circumference of the trunk of a tree which is 100 years old, [1]
- (c)** the age of the tree where the circumference of the trunk is 85 cm. [2]
- (v)** Comment briefly on the reliability of the estimates in part **(iv)**. [3]

END OF PAPER